A wire, when bent in the form of a square, encloses an area of 121cm square. If the same wire is bent to form a circle, find the area of the circle.

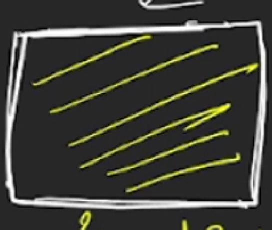
A.144 B.180 C.154 D.186

Concept :- circum

$4a = 2 \times \frac{22}{7} \times r$

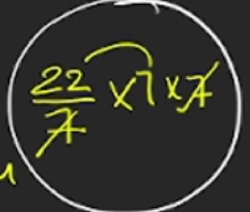
$\cancel{4} \times 11 = \cancel{2} \times \frac{\cancel{22}}{7} \times r$

$\boxed{r = 7\ cm}$



$a^2 = 121$

$\boxed{a = 11}$



$\frac{22}{7} \times 7 \times 7$

Area = ?

04:50 -07:34

The height and radius of a cylinder are in the ratio 7:3. The five-eleventh part of the vessel is full of 2.43 liters of water. What is the curved surface area of the vessel? (Take π= 22/7)

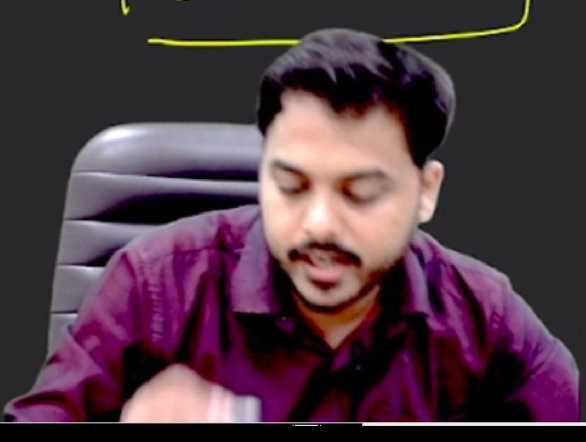
1. 1292 sq.cm
2. 1326 sq.cm
3. 1248 sq.cm
4. 1188 sq.cm

$C.S.A = 2 \times \frac{22}{7} \times 3 \times \frac{3}{10} \times 7 \times \frac{3}{10} \times 100$

$= 1188\ cm^2$ Ans.

$x = \frac{3}{10}$

$\boxed{1\ lit = 100\ cm^2}$



$x^3 = \frac{27}{1000} =$

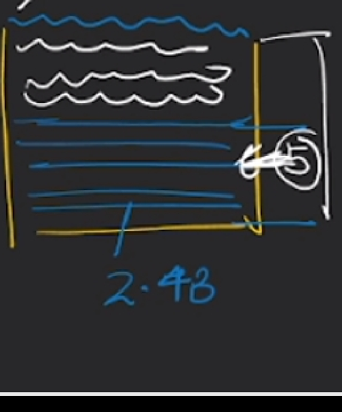
(Vol)

$h = 7x$

$r = 3x$

$\frac{22}{\cancel{7}} \times 3x \times 3x \times \cancel{7}x = \frac{2.43 \times 11}{5}$

$x^3 = \frac{\cancel{243}^{27} \times \cancel{11}}{5 \times 100 \times \cancel{22}_{2} \times \cancel{9}}$



2.43

Q1. The length, width, and height ratio of a cuboid is 7:5:4 and its entire surface area is 2656 square cm. Find its width.

a. 16 cm b. 20 cm c. 28 cm d. 300

$S.A = 2(lb + bh + hl)$

$2656 = 2(35x^2 + 20x^2 + 28x^2)$

$x^2 = \frac{2656}{2 \times 83}$

$x^2 = 16$ $\boxed{x = 4}$

$5 \times 4 = 20\ cm$

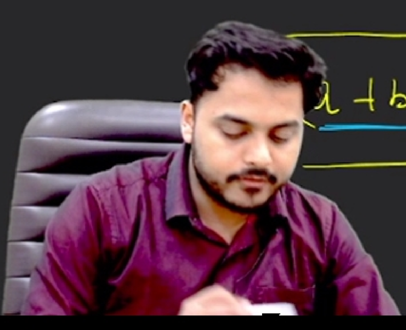
Q2. The sum of the length, width and height of a cuboid x cm and the length of its diagonal is y cm, then its total surface area will be-

a. $x^2 - y^2$ b. $x^2 + y^2$ c. $\frac{1}{2}(x^2 - y^2)$ d. $(x-y)^2$

$l + b + h = x\ cm$

$y^2 = l^2 + b^2 + h^2 \leftarrow y = \sqrt{l^2 + b^2 + h^2}$

$T.S.A = 2(lb + bh + hl)$



$(a + b + c)^2 = \boxed{a^2 + b^2 + c^2} + \boxed{2(ab + bc + ca)}$

$x^2 = y^2 + T.S.A$

$h\ 2(l+b)$ $l:b$

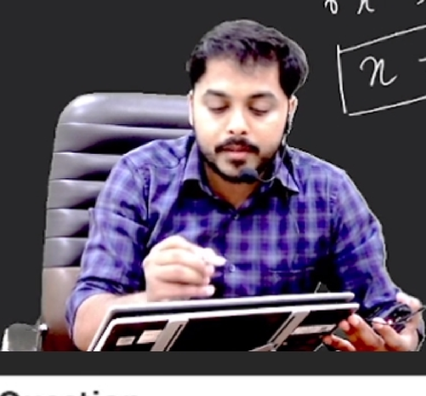
Q3. The length and breadth ratio of an auditorium is 3: 2. If the roof area is 216 square meters and the area of the four walls is 480 square meters, then what will be the height of the auditorium?

a. 6 m b. 10 m c. 8 m d. 12 m

$l \times b = 216$

$6x^2 = 216$

$\boxed{x = 6}$



$2h(l + b) = 480$

$\cancel{2} \times h(\cancel{30}) = \cancel{480}$

$\boxed{h = 8\ mtrs}$

Question

The area of the four walls of a room is $108\ m^2$. If the height and length of the room are in the ratio of $2:5$ and the height and breadth are in the ratio $4:5$, then the area, in m^2, of the floor of the room is

$l \times b = 50x^2 = 50 \times \frac{9}{10} = 45$

$2(l + b)\ h = 108$

$2(\cancel{15}^{5}x) \times \cancel{4}x = \cancel{108}^{27\ 9}$

$\boxed{x^2 = \frac{9}{10}}$

$\frac{h}{l} = \frac{2 \times 2}{5 \times 2}$ $\frac{h}{b} = \frac{4}{5}$

$h : l : b$

$4 : 10 : 5$

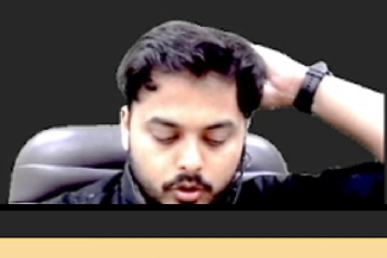
Q.

Find the total surface area (in cm^2) of a right circular cylinder of diameter 7 cm and height 6 cm.

a. 187 b. 209 c. 163 d. 149

$TSA = 2\pi rh + 2\pi r^2 = 2\pi r(r + h)$

$= \cancel{2} \times \frac{22}{\cancel{7}} \times \frac{\cancel{7}}{\cancel{2}} \left(\frac{7}{2} + 6\right)$

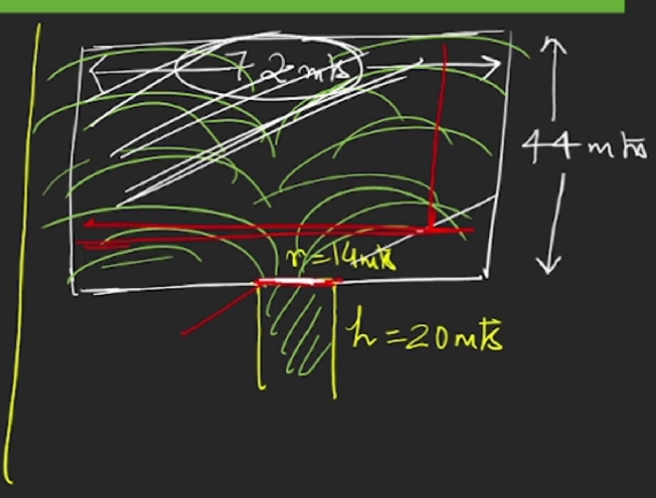


$= 11 \times 19 = 209$

Q. A cylindrical well of height 20 metres and radius 14 metres is dug in a field 72 metres long and 44 metres wide. The earth taken out is spread evenly on the field. What is the increase (in metre) in the level of the field?

a. 6.67 b. 3.56 c. 5.61 d. 4.83

$h = \frac{22 \times 2 \times 14 \times 20}{72 \times 44 - \frac{22}{7} \times 14 \times 14}$



Campusmonk

Question :

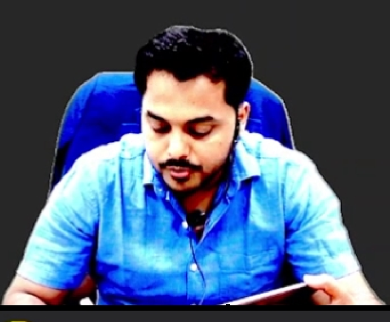
A solid cone of height 24 cm and radius of its base 8 cm is melted to form a solid cylinder of radius 6 cm and height 6 cm. In the whole process what percent of material is wasted?

a. ~~48.5~~ b. ~~37.5~~ c. 57.8 d. ~~64~~

$\frac{\text{Cone Vol}}{\text{Cy Vol}} = \frac{\frac{1}{\cancel{3}} \cancel{\pi} (8)^2 \times \cancel{24}^{8\ 4}}{\cancel{\pi} (6)^2 \times \cancel{6}_{3}} = \frac{64}{27}$

$64 \longrightarrow 27$

$\Rightarrow \boxed{\frac{37}{64} \times 100}$



$32 \xleftarrow{50\%}$

%

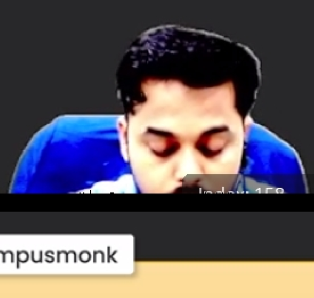
Campusmonk

Question: Three spherical balls of radius 2 cm, 4 cm and 6 cm are melted to form a new spherical ball. In this process there is a loss of 25% of the material. What is the radius (in cm) of the new ball?

a. 6 b. 8 c. 12 d. 16

$\boxed{(V)_{New} = \frac{3}{4} \times (V)_{original}}$

$(r)^3 = 216$



$\boxed{r = 6}$

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Question:

Three spherical balls of radius 3 cm, 2 cm and 1 cm are melted to form a new spherical ball. In this process there is a loss of 25% of the material. What is the radius (in cm) of the new ball?

a. 5 b. 3 c. 6 d. 8

$V_{new} = \frac{3}{4} V_{original}$

$\frac{4}{3}\pi(r)^3 = \frac{3}{4} \times \frac{4}{3}\pi(27 + 8 + 1)$